

3 Methodology and Data

3.1 Methodology

3.1.1 Output Gap Estimation

To estimate output gaps, we employ a total of thirteen output gap estimation methodologies. All output gap measures are standard in the literature, and most of them are straightforward to implement in standard statistical or econometric software. We intentionally use such off-the-shelf approaches to address the model uncertainty associated with the most popular output gap measures.

The first ten gaps are estimated from quarterly real GDP data. We begin with the [Hodrick and Prescott \(1981\)](#) (HP) filter, the most commonly used univariate filter in the literature, and two band-pass filters: [Baxter and King \(1999\)](#) and [Christiano and Fitzgerald \(1999\)](#). We also employ a modified Beveridge–Nelson decomposition following [Kamber et al. \(2018, 2025\)](#), the original [Hamilton \(2018\)](#) filter and its modification by [Quast and Wolters \(2022\)](#), and a UC-0 unobserved components model following [Harvey \(1985\)](#) and [Clark \(1987\)](#). Finally, we estimate three structural VAR (SVAR) models with long-run restrictions in the spirit of [Blanchard and Quah \(1989\)](#), each including quarterly real GDP growth alongside one of the following variables: the employment rate, the unemployment rate, or the inflation rate. In each SVAR, potential output is recovered as the level of output that would prevail in the absence of demand shocks using historical decomposition.⁵

The remaining three gaps are taken from the official estimates of the European Commission, the OECD, and the IMF. The European Commission and the OECD both rely on a Cobb–Douglas production function approach, estimating potential output via the natural rate of unemployment at which unit labor costs (European Commission) or core inflation (OECD) are stable; full methodological details are provided in [Havik et al. \(2014\)](#) and [Chalaux and Guillemette \(2019\)](#), respectively. The IMF does not apply a single unified methodology across countries; instead, analysts select the approach considered most appropriate for each country, drawing on univariate and multivariate filters, the production function, and expert judgment ([De Masi, 1997](#); [Barkema et al., 2020](#)).

⁵For further details on each estimation method, see Appendix [A](#).

Since the institutional estimates are published at the annual frequency while the remaining ten gaps are quarterly, we convert the annual institutional gaps to a quarterly frequency. Specifically, we first recover annual potential GDP implied by each institution’s gap estimate, then interpolate it to quarterly frequency using a quadratic-match sum procedure, which preserves the annual sum of potential output and ensures the interpolated series is consistent with the original institutional estimates. The quarterly output gap is then computed as the percentage difference between actual seasonally adjusted real GDP and the interpolated quarterly potential GDP. This yields thirteen output gap series used throughout the analysis.

3.1.2 Quasi-real-time and final output gaps

Following [Orphanides and Norden \(2000\)](#), we estimate and use both the quasi-real-time and final output gaps. The quasi-real-time output gap estimate for period t is based on time series which as the last data point includes the data for period t (the expanding window method). That is, one attempts to historically reconstruct the gap estimate that could have been obtained for period t right after all the data required to estimate the gap for period t have been published. The data for later periods (after period t) are not used for this estimation because they were not available in real-time when it was first possible to estimate the gap for period t . The difference between real-time and quasi-real-time estimates is that real-time estimates use original (vintage) data, while quasi-real-time estimates use the most recent (currently available) vintages of data, which were possibly revised compared to the vintage data (for example, GDP figures get revised often, etc.). Both quasi-real-time and real-time output gap estimates are essentially obtained using the expanding window method. For example, if one is trying to estimate the (quasi) real-time output gap for France for 2012Q3, the last data used for the estimation is the data for 2012Q3.

On the other hand, final estimates of the output gap use all currently available data to estimate the gap for period t . For example, if one is trying to estimate the output gap for France for 2012Q3, all of the currently available data will be used to estimate that gap, for instance, the data up until 2023Q4.

3.1.3 Model uncertainty and model averaged output gap (MAOG)

In total we employ up to 13 different methodologies to estimate the K output gaps c_{itk} for a specific country.⁶ Due to data availability and the credibility of the estimated gaps in some countries we use smaller number of gaps $K < 13$.

The total number of models employed can be much higher, [Morley and Panovska \(2020\)](#) for example use 13 linear and 14 nonlinear models in order to estimate MAOG. In this paper, we use 13 methodologies, but we can easily expand the model along that dimension.

Since we are focused on output gaps standard deviation, we use only symmetric, or linear output gaps. The literature also recognizes an important role of asymmetric or nonlinear output gaps ([Morley and Piger, 2012](#); [Morley and Panovska, 2020](#); [Arčabić et al., 2024](#)). However, combinations of both symmetric and asymmetric output gaps would increase the dispersion by construction, and thus affect the measured standard deviation. Therefore, in this paper we do not pursue this avenue.

After K gaps have been estimated, typically different averaging techniques are employed to calculate a single estimate of the model-averaged output gap c_{it} . Authors usually use the unweighted average ([Morley and Panovska, 2020](#)), weighted average based on the Bayesian approach ([Morley and Piger, 2012](#)), or a theoretical justification such as the Phillips curve relationship vis a vis the inflation rate ([Arčabić et al., 2024](#)).

In the paper we estimate the model average output gap (MAOG) as the equally weighted average of K output gaps for each country and time period:

$$c_{it} = \frac{1}{K} \sum_{k=1}^K c_{itk} \quad (1)$$

where $k = 1...K$ number of employed output gap methodologies, and c_{itk} is the estimated MAOG in time t and country i .

Although the calculation and robustness of MAOG approach has been extensively researched in the literature, as far as authors are aware the level of model uncertainty is rarely employed in the analysis of output gap. Formally, we define uncertainty as the standard

⁶HP filter, Hamilton filter, Quast and Wolters filter, Baxter-King filter, Christiano-Fitzgerald filter, Kamber, Morley and Wong's refined BN filter, UC-0 method, three SVAR models and the output gap estimates from European Commission, OECD and IMF.

deviation of estimated gaps c_{itk} vis-à-vis the MAOG c_{it} .

$$\delta_{it} = \sqrt{\frac{\sum_1^K (c_{itk} - c_{it})^2}{K}} \quad (2)$$

Every estimate of the MAOG c_{it} has a different level of uncertainty (standard deviation) in each country and time period, denoted as δ_{it} .

3.1.4 Linear and threshold dynamic panel model

Many authors are well aware that model uncertainty exists and that it can be quite volatile through time and space, but the key question in this paper is how to utilize the information about the level of uncertainty in the process of gap estimation and eventually in policy making. In this paper, we analyze the connection between GDP growth rate, MAOG, and model uncertainty. We employ linear and nonlinear dynamic panel model ([Chan, 1993](#)) in order to investigate the issue.

We start with one way and two way linear dynamic fixed effect panel model:

$$\delta_{it} = \alpha_i + \gamma_t + \rho\delta_{it-\tau} + \beta\Delta y_{i,t} + e_{1i,t} \quad (3)$$

and proceed with the following form of the threshold two way dynamic panel data model

$$\delta_{it} = \alpha_i + \gamma_t + \begin{cases} \rho_1\delta_{it-\tau} + \beta_1\Delta y_{i,t} + e_{1i,t} & \text{if } q_{i,t-d} \leq \theta \\ \rho_2\delta_{it-\tau} + \beta_2\Delta y_{i,t} + e_{2i,t} & \text{if } q_{i,t-d} > \theta, \end{cases} \quad (4)$$

where δ_{it} is the standard deviation between various gap estimates (a measure of uncertainty), τ is the lag of dependent variable, we estimate model without lag dependent variable, with $\tau = 4$ and with two lag dependent variables $\tau = 4$ and $\tau = 1$; $\Delta y_{i,t}$ denotes the GDP growth rate where y stands for $\log(\text{GDP})$, $q_{i,t}$ is the threshold variable. We estimate model with GDP growth rate, MAOG and absolute value of GDP growth rate as threshold variables. θ is the estimated threshold value/parameter; α_i and γ_t capture cross-sectional and time fixed effects, respectively; and d is the delay parameter. We use both $d = 0$ and $d = 1$ to address

the problem of intertemporal links between regimes.

In order to estimate threshold value θ we grid search for the best sample split by minimizing loss function represented by root mean square error parameter. The grid search for the threshold value is performed through all the observations of threshold variable between 10th and 90th percentile.⁷

3.1.5 Probit models and recession forecasting

We also use the output gap standard deviation ($\delta_{i,t}$) to forecast recessions. Recessions are identified using a standard rule of thumb, where a recession is called if the Y-o-Y growth rate of real GDP is negative for two consecutive quarters. This binary indicator (1 for recession quarters, and 0 otherwise) serves as the dependent variable in all models estimated below.

We use panel probit models on a balanced panel of 25 EU member states plus the United Kingdom observed at quarterly frequency. The full estimation sample begins in 2002:Q1; the out-of-sample forecast evaluation window covers 2012:Q1–2024:Q4, where the cutoff is chosen to leave a sufficient estimation window that includes the Global Financial Crisis while preserving more than a decade of genuine out-of-sample observations spanning the euro-area sovereign debt crisis, the COVID-19 recession, and the 2022–2023 energy-price shock.

The baseline specification is a panel probit with an unobserved country effect α_i

$$P(R_{i,t+h} = 1 \mid \mathbf{x}_{i,t}, \alpha_i) = \Phi(\mathbf{x}_{i,t}'\boldsymbol{\beta} + \alpha_i), \quad (5)$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function, $\boldsymbol{\beta}$ is the coefficient vector, and the forecast horizon is $h \in \{1, 2, 3, 4\}$ quarters. Forecasts are produced directly rather than by iteration, so a separate model is estimated for each h . The regressor vector $\mathbf{x}_{i,t}$ contains an intercept and the output gap model uncertainty measure $\delta_{i,t}$ as its only non-constant element. In the static specification $\mathbf{x}_{i,t} = (1, \delta_{i,t})'$; in the dynamic specification the lagged recession indicator $R_{i,t}$ is added to capture state persistence.

We estimate equation (5) under two treatments of α_i . In the random effects (RE) probit, $\alpha_i \sim \mathcal{N}(0, \sigma_\alpha^2)$ is assumed independent of $\mathbf{x}_{i,t}$ and the likelihood is evaluated by adaptive

⁷For the robustness check we also perform grid search between 5th and 95th percentile (see Figures 5 and B.1 for details).

Gauss–Hermite quadrature. In the fixed effects (FE) specification, we estimate a pooled probit with a full set of country indicator variables. We refer to this as an "FE probit" for brevity, but note that with finite T the unit-dummy MLE suffers from the incidental parameters problem (Neyman and Scott, 1948): the slope estimator is asymptotically biased of order $O(1/T)$ and the bias does not vanish under fixed- T asymptotics (Greene, 2004). With $T \approx 92$ quarterly observations per country the bias is expected to be small. Inference throughout uses standard errors clustered at the country level, which is robust to arbitrary within-country serial correlation and to the conditional heteroskedasticity induced by the probit link. Combining the two static and two dynamic specifications yields four models: RE-static, RE-dynamic, FE-static, and FE-dynamic.

In the RE-dynamic specification the lagged outcome $R_{i,t}$ is by construction correlated with α_i , giving rise to the initial conditions problem (Heckman, 1981). We address this using the auxiliary-model approach of Wooldridge (2005), which parameterizes the conditional distribution of the unobserved effect as

$$\alpha_i \mid R_{i,1}, \bar{\delta}_i \sim \mathcal{N}(a_0 + a_1 R_{i,1} + a_2 \bar{\delta}_i, \sigma_a^2), \quad (6)$$

where $R_{i,1}$ is the initial recession state and $\bar{\delta}_i$ is the within-country time-series mean of the uncertainty regressor. Substituting into (5) and integrating out α_i yields a standard RE probit in which $R_{i,1}$ and $\bar{\delta}_i$ enter as additional regressors, so that the unobserved effect is modeled conditional on the initial state rather than assumed independent of the dynamic history. The static RE specification contains no lagged dependent variable and is therefore free of the initial conditions problem.

Our probit models are deliberately simple, because the idea is not performing a forecasting horse race and finding the best model, but showing that model uncertainty has properties that are useful in forecasting recessions. The choice of probit model is not by chance. Chalaux et al. (2025) show that an algorithm that selects appropriate probit models, matches the performance of Random Forests in out-of-sample predictions, including the period of the Global Financial Crisis. Such performance is attributed to a *wisdom of crowds* property, as predictions from probit equations are averaged.

Out-of-sample forecasting is based on a recursive expanding window scheme. For each

quarter t from 2012:Q1 to 2024:Q4, the model is re-estimated using all panel observations up to but excluding quarter t , and the resulting coefficient estimates are used to compute the predicted recession probability for quarter t . The initial training sample thus spans from 2002:Q1 to 2011:Q4, and it expands by one quarter at a time. This recursive procedure replicates the information constraints of a real-time forecasting exercise, since each forecast uses only data that would have been available at the time it was produced.

The second part of the analysis estimates a probit model for each of the 26 countries, in a time-series setting. As in the panel case, static and dynamic probit specifications are estimated with lags from one to four quarters. In the time series setting, there are no fixed effects, and therefore, there is no potential bias as in a panel setting.

Model forecasting performance is evaluated using the area under the receiver operating characteristic curve, AUROC. The ROC curve plots the true positive rate against the false positive rate as the classification threshold for the predicted recession probability is varied continuously from zero to one, and the AUROC is the integral of this curve. An AUROC of 0.5 indicates no discriminatory power relative to a random walk model, values around 0.7 are generally considered acceptable, values around 0.8 are considered good, and values above 0.9 indicate excellent discrimination. To assess whether differences in AUROC values across different specifications are statistically significant, we use the DeLong test ([DeLong et al., 1988](#)). The DeLong test compares two correlated ROC curves estimated on the same observations and tests the null hypothesis that their AUROC values are equal.

3.2 Data

Since every output gap estimation method requires data for the real GDP of the country whose gap is being estimated, data on seasonally and calendar-adjusted real GDP (in euros) was taken from two databases, the Eurostat database ([Eurostat, 2025c](#)) and the OECD database ([OECD, 2025b](#)). The frequency of the data is quarterly, and the reason why these two databases are combined is to obtain the longest possible series for all countries in the sample. The real quarterly GDP series were expressed in logarithmic form before applying each univariate method. The difference between the logarithms of actual and potential GDP represents the output gap. Real GDP data for the 26 countries in our sample is shown in

Figure A.1, and Figure A.2 displays the real GDP growth rates.

A similar approach was used when retrieving the data on the unemployment rate and the employment rate that are needed to estimate the output gap using the Blanchard-Quah SVAR model. To obtain the longest possible series of data on the unemployment rate, for some countries, we use quarterly data for the age group 15+ from the OECD database (OECD, 2025c), while for the other countries we use the data for the age group 15-74 obtained from the Eurostat database (Eurostat, 2025e) and the Eurostat database with historical data (Eurostat, 2025f). The use of different age groups does not affect our results because our research is not based on mutual comparison of gaps between countries. Quarterly data on the employment rate in the age group 15-64 was taken from Eurostat, combining current (Eurostat, 2025a) and historical (Eurostat, 2025b) data depending on the availability of data for each country in the sample. The data for the UK's employment rate was obtained from the OECD's database (OECD, 2025c). The quarterly GDP deflator (price) data needed to estimate the output gap using the Bayoumi and Eichengreen SVAR model was taken from Eurostat (Eurostat, 2025d) and the UK's Office for National Statistics (Office for National Statistics, 2025). The employment and unemployment rate data is shown in figure A.3, while figure A.2 displays the GDP deflator rate of inflation.

When it comes to real-time gaps estimated by the European Commission, the data was taken from the autumn editions of the publication "Economic Forecast" published in the period from 2003 to 2025, from which we obtained the values of the gaps for the period from 2002 to 2024 (European Commission (2025)). We take the first publicly available output gap estimate as the real-time gap estimated by the European Commission. The first year for which the European Commission published gap estimates was 2002 for a wider set of European Union countries, hence this year was taken as the starting year. Final gaps estimated by the European Commission were obtained from the same publication and from the last vintage which published output gap estimates for a specific year. A similar approach was used to obtain the real-time and final output gaps estimated by the IMF and the OECD. The IMF publishes gap estimates as a part of the "World Economic Outlook" publication, and these gap estimates are available in the IMF database (IMF, 2025). Estimates of output gaps published by the OECD were taken from the OECD database and the publication

”Economic Outlook” ([OECD, 2025a](#)).

4 Results

The results section is divided into three parts. First, we present the results of output gaps estimation and model uncertainty. Secondly, we provide estimates of linear and nonlinear dynamic panel models. Third, we perform different forecasting exercises, showing that the model uncertainty is a meaningful signal in recession forecasting, and that it includes a valuable information.

4.1 Output gaps and model uncertainty

Figure 1 presents all estimated gaps used in the analysis for each country. All depicted gaps are estimated for the quasi-real-time sample.⁸ As shown in Figure 1, output gaps are similar, but there are periods when the divergence between various methodologies is large and persistent. In a few countries, such as the UK, output gap measures are highly synchronized. But in the most countries, there are substantial disagreements between output gap measures, even about the current state of the economy.

To see if such disagreements are random or systematic, we calculate the model uncertainty measured by the standard deviation. Results for the quasi-real-time sample are presented in Figure 2. The Figure shows three versions of standard deviation based on the quasi-real time. The first includes all thirteen output gaps, the second excludes institutional output gaps, while the third excludes [Hamilton \(2018\)](#) filter and UC-0 output gap.⁹ Periods of negative Y-o-Y GDP growth are shaded.

In many countries in our sample, shaded periods are usually periods of higher model uncertainty. Three major crisis are common to almost all the countries: the Global Financial Crisis (GFC) of 2008-2010, the Sovereign debt crisis in the Eurozone (2010-2012) and the Covid recession of 2020-2021. A substantial increase in the model uncertainty is present

⁸Due to the differences in data availability, MAOG was calculated using a smaller number of gaps (fewer than 13) for some countries. Full sample output gap results are presented in Figure C.1.

⁹Figure C.2 presents the comparison between model uncertainty estimated for quasi-real-time gaps and final gaps.